Smart Nutrition

Data Engineering & Exploratory Data Analysis

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Contents

1 1. Data Source & Context **2**
1.1 Project Overview 2
1.2 Internal Dataset 2

2 2. Data Engineering **2**
2.1 Feature 1: BMI Categorization 2
2.2 Feature 2: Health Risk Score 3
2.3 Feature 3: Caloric Balance 3

3 3. Exploratory Data Analysis (EDA) **3**
3.1 3.1 Data Quality & Missing Values 4
3.2 3.2 Target Distribution (Classification) 4
3.3 3.3 Impact of BMI on Diet Choice 5
3.4 3.4 Nutritional Correlations (Regression Targets) 6
3.5 Health Risk vs. Diet Plan 7

4 Conclusion **7**
4.1 4.1 Achievements 7
4.2 4.2 Next Steps (Phase 2) 7

1 1. Data Source & Context

1.1 Project Overview

The objective of this project is to build an intelligent system that recommends personalized diet plans and nutritional targets based on a user's medical history, demographics, and lifestyle. This phase focuses on transforming raw medical records into structured, enriched features suitable for Machine Learning.

1.2 Internal Dataset

The primary data source is `Personalized_Diet_Recommendations.csv`, containing the following key features:

- **Target Variables:**
  - *Classification:* `Recommended_Meal_Plan` (e.g., Balanced, High-Protein, Low-Carb).
  - *Regression:* `Recommended_Calories`, `Protein`, `Carbs`, `Fats`.
- **Demographics:** `Age`, `Gender`, `Height_cm`, `Weight_kg`.
- **Medical History:** `Chronic_Disease`, `Blood_Pressure`, `Blood_Sugar`, `Cholesterol`.
- **Lifestyle:** `Daily_Steps`, `Exercise_Frequency`, `Sleep_Hours`.

2 2. Data Engineering

Since external web scraping was not required for this dataset, we focused on **Internal Feature Engineering**. We derived three critical features to provide clinical context to the models.

2.1 Feature 1: BMI Categorization

We converted the continuous `BMI` (Body Mass Index) into clinical categories to simplify pattern recognition.

```python
def get_bmi_category(bmi):
    if bmi < 18.5:
        return 'Underweight'
    elif bmi < 25:
        return 'Normal'
    elif bmi < 30:
        return 'Overweight'
    else:
        return 'Obese'

# Applied to dataset
df['bmi_category'] = df['BMI'].apply(get_bmi_category)
```

Listing 1: BMI Categorization Logic

Purpose: This feature helps the model distinguish dietary needs (e.g., High-Calorie for Underweight vs. Low-Carb for Obese).

2.2 Feature 2: Health Risk Score

We synthesized a composite score to quantify the overall health risk of the patient.

```
def calculate_health_risk(row):
    score = 0
    if row['Blood_Pressure_Systolic'] > 140: score += 1
    if row['Cholesterol_Level'] > 200: score += 1
    if row['Blood_Sugar_Level'] > 126: score += 1
    if row['Smoking_Habit'] == 'Yes': score += 1
    if row['Chronic_Disease'] != 'None': score += 1
    return score

df['health_risk_score'] = df.apply(calculate_health_risk, axis=1)
```

Listing 2: Health Risk Calculation

Purpose: A higher score indicates a need for stricter dietary control (e.g., Low-Salt, Low-Sugar).

2.3 Feature 3: Caloric Balance

We calculated the difference between the user's actual intake and the recommended target.

```
df['caloric_balance'] = df['Caloric_Intake'] - df['Recommended_Calories']
```

Listing 3: Caloric Balance Logic

Purpose:

- *Positive Balance:* User is overeating (Surplus).
- *Negative Balance:* User is undereating (Deficit).

This helps the model understand current adherence levels.

The result is an enriched dataset saved as `diet_data_enriched.csv`.

3 3. Exploratory Data Analysis (EDA)

With the enriched dataset, we perform visual analysis to understand the distribution of diet plans and the relationship between health indicators and nutritional targets.

3.1 3.1 Data Quality & Missing Values

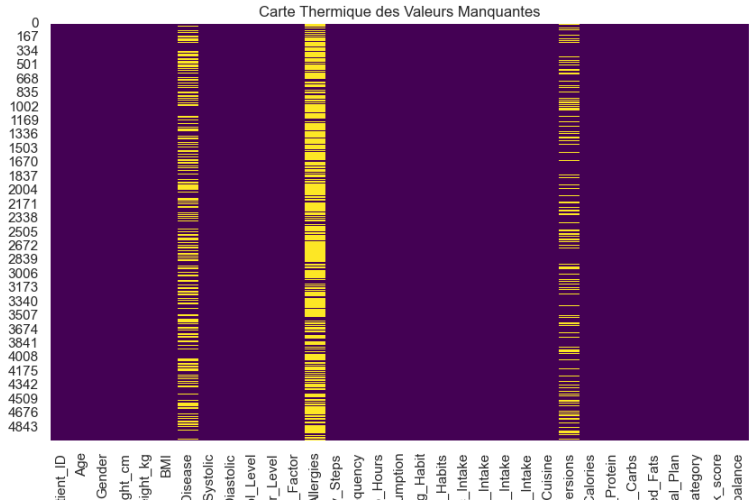

Figure 1: Missing Values Heatmap

Analysis: Figure 1 reveals missing data in several columns (e.g., `Allergies`, `Daily_Steps`).
Implication for Week 2: We will utilize `SimpleImputer` within a Scikit-Learn Pipeline (Median for numbers, Most-Frequent for categories) to handle these gaps.

3.2 3.2 Target Distribution (Classification)

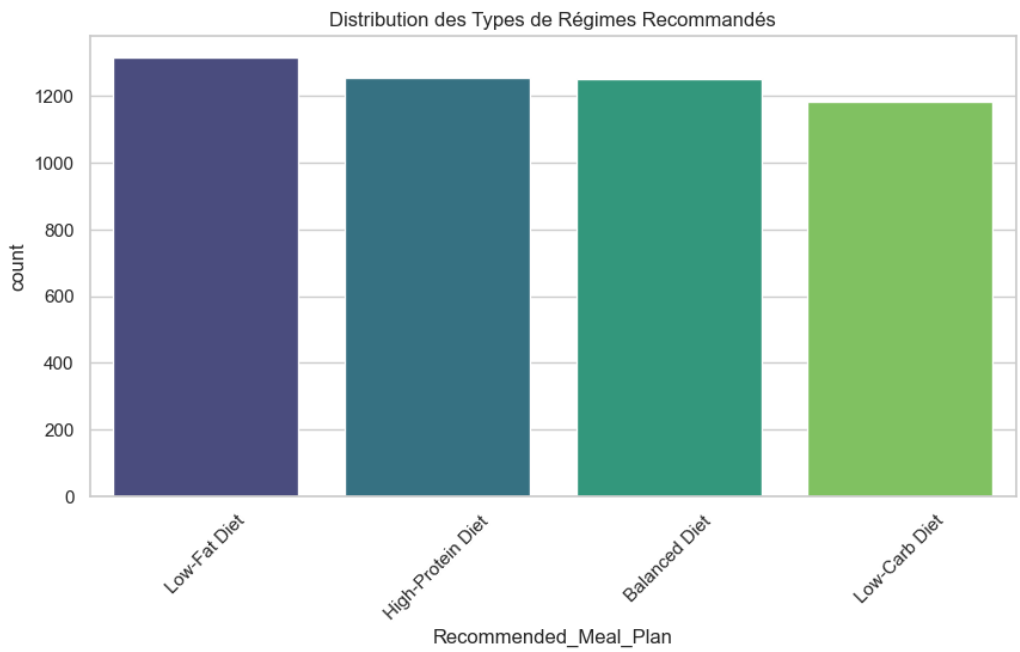

Figure 2: Distribution of Recommended Meal Plans

Analysis: Figure 2 shows the frequency of recommended diet plans.
Implication for Week 2: We check for class imbalance. If "Balanced Diet" is dominant, we may need to use **SMOTE** to prevent the model from biasing towards the majority class.

3.3 3.3 Impact of BMI on Diet Choice

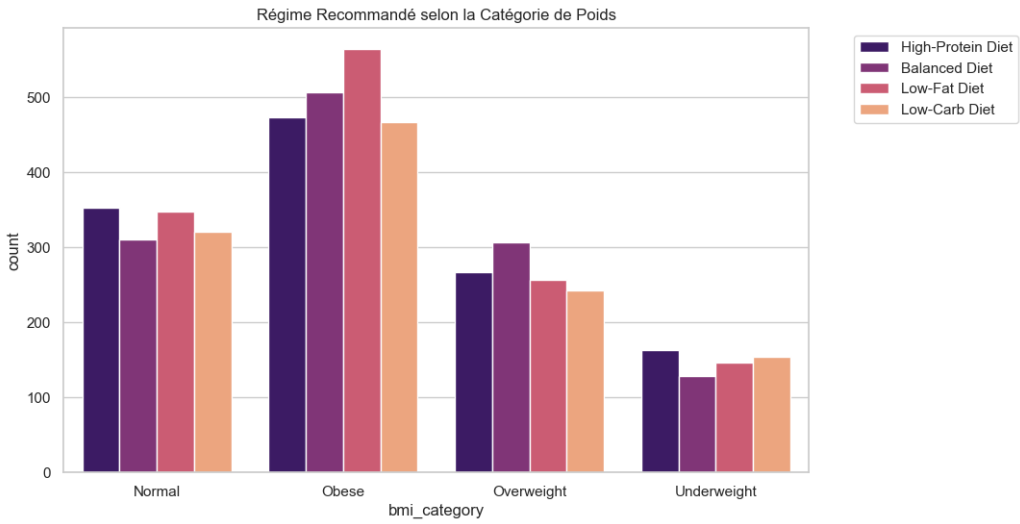

Figure 3: BMI Category vs. Recommended Diet

Analysis: Figure 3 visualizes the clinical logic of the recommendations.

- We expect "Obese" or "Overweight" categories to correlate strongly with "Low-Fat" or "Low-Carb" diets.
- "Underweight" users might be directed towards "High-Calorie" or "High-Protein" diets.

3.4 3.4 Nutritional Correlations (Regression Targets)

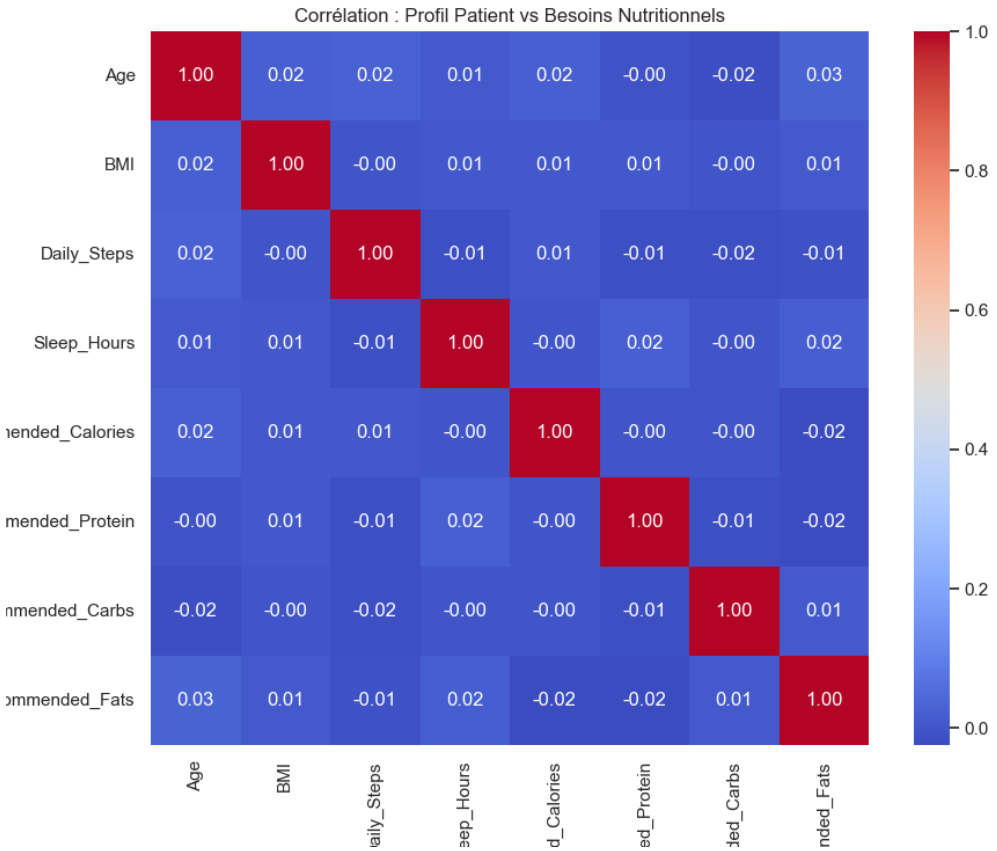

Figure 4: Correlation Matrix: Demographics vs. Nutrition Targets

Analysis: Figure 4 highlights the relationships between physical attributes and nutritional needs.

- We check correlation between `Weight_kg` and `Recommended_Calories`.
- We check correlation between `Exercise_Frequency` and `Recommended_Protein`.

High correlations validate the use of Linear Regression or Random Forest for predicting these numeric targets.

3.5 Health Risk vs. Diet Plan

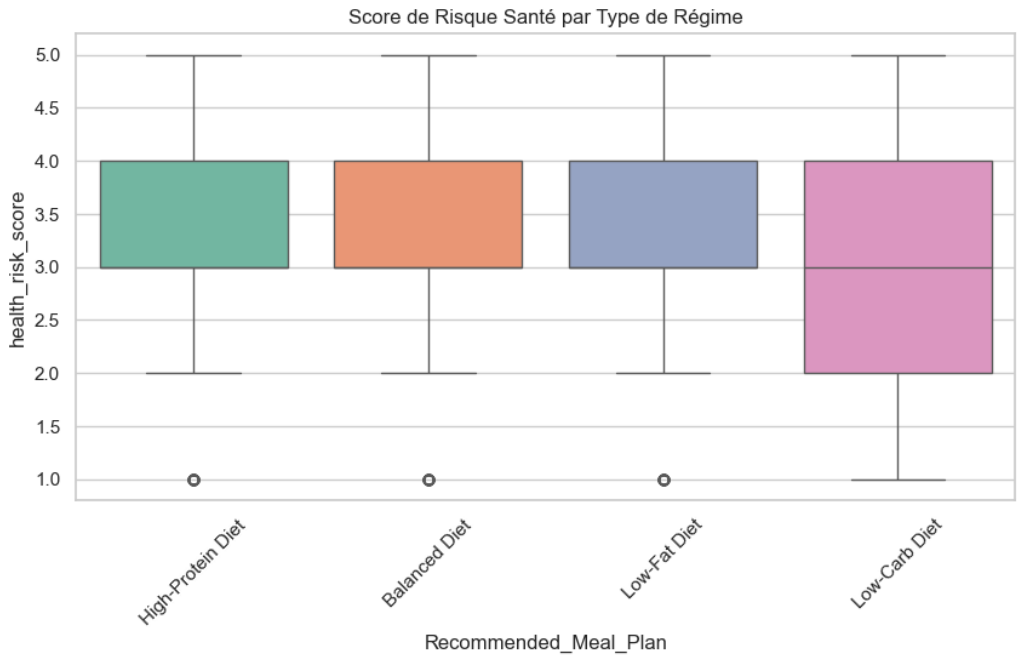

Figure 5: Health Risk Score vs. Diet Plan

Analysis: Figure 5 indicates whether stricter diets (e.g., Low-Sodium) are assigned to patients with higher `health_risk_score`.

4 Conclusion

Phase 1 has successfully transformed raw medical data into an enriched dataset ready for advanced Machine Learning techniques.

4.1 4.1 Achievements

1. **Data Engineering:** Created three critical features:
   - `bmi_category`: Clinical segmentation of weight status.
   - `health_risk_score`: A composite metric for patient health severity.
   - `caloric_balance`: Insight into current eating habits vs. needs.
2. **Insights:** EDA revealed distinct patterns where BMI and Health Risk Score significantly influence the choice of diet plan (Classification) and nutritional targets (Regression).

4.2 4.2 Next Steps (Phase 2)

The groundwork is now laid for constructing a robust Machine Learning Pipeline in Phase 2, which will involve:

- **Model 1 (Regression):** A Multi-Output Regressor to predict `Recommended_Calories`, `Protein`, `Carbs`, and `Fats`.

- **Model 2 (Classification):** A Multi-Class Classifier to predict the `Recommended_Meal_Plan` (e.g., Balanced vs. Keto).
- **Optimization:** Utilizing **GridSearchCV** to find optimal hyperparameters.
- **Tracking:** Logging experiments for both models using **MLflow**.
- **Integration:** Connecting these models to an AI API (LLM) to generate detailed 7-day meal plans based on the predictions.